Cartesian Impedance Controller Analysis on the Z1 Unitree Manipulator

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***Abstract*—This report covers the derivation and implementation of the impedance controller to study its performance under different environmental conditions and control parameters. This is to study force-based controllers widely leveraged in human-robot interaction tasks to adhere to safety requirements based on force and compliance. This study implements a simplified Cartesian impedance controller on the 6 degree-of-freedom (DOF) Unitree Z1 manipulator in simulation. The purpose is to evaluate the tracking behaviour and interaction forces under 3 scenarios (free motion, wall contact, and external disturbance) under varying Cartesian stiffness. The experimental results illustrate that increasing the stiffness results in increased positional accuracy but requires greater exerted forces from the manipulator. On the physical robot, this can lead to damage on itself and/or objects/humans it comes in contact with as a trade off. Therefore, there must be a balance between objectives to enforce safety with meaningful production throughput. The code and video of the experiments can be found here github.com/chungtyler/z1-impedance-control.**

I. Introduction

Traditional industrial robotic manipulators have been proven useful for several years, automating menial tasks that require high speeds and accuracy. This technology has been used extensively in the automotive, agricultural, and medical industries, from heavy-duty manoeuvring to pick-and-place operations. The limitation of position-based robots is that they lack the capability to comply with the environment, ignoring people during execution [1]. This makes such technologies inherently dangerous to workers requiring safety barriers by isolating people from the workspace. Even with additional layers of safety accidental events can occur in the workspace since the robot itself isn't aware of the person [1]. The introduction of collaborative robots (CoBots) aims to not only mitigate this issue but also promote cohesion between humans and robots for shared tasks. Several layers of safety are considered, from physical design to low-level control. This enables human-aware operation, ensuring safety limits are imposed while benefiting from both generalizable human capabilities with robot repeatability for collaborative tasks.

This reports aims to investigate the impedance controller, a fundamental force-based controller used in CoBots due to its inherent design for compliance and force limiting. The performance will be characterized to develop an intuition behind its behaviour and parameter sensitivity under different environmental conditions.

II. Impedance Control

The impedance controller is a force-based controller used to simulate the dynamics of a virtual mass-spring-damper system in Cartesian space [2]. The overall goal is to relate the accelerations between the systems and solve for the desired Cartesian force. The derivation and notation are with respect to [3].

A. Flexible Joint Dynamics

The robot manipulator assumes the form of the flexible joint dynamics model for n-DOF [3].

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau_d + \tau_{ext} \tag{1}$$

Where $q \in \mathbb{R}^n$ is the vector of n-joint positions, $M(q)$ is the joint space inertia matrix for each link, $C(q,\dot{q})$ the Coriolis and centrifugal torques, $g(q)$ the gravity torques, τ_d the desired joint torques, and τ_{ext} are the external joint torques. Equation 1 is then isolated for $\ddot{q}$ which results in

$$\ddot{q} = M(q)^{-1}[\tau_d + \tau_{ext} - C(q,\dot{q})\dot{q} - g(q)] \tag{2}$$

To relate back to the mass-spring-damper model, $\ddot{q}$, τ_d, and τ_{ext} are converted into Cartesian space. This is achieved through the spatial Jacobian $J(q) \in \mathbb{R}^{6\times n}$ joint space mapping and the concept of virtual work $J(q)^T$. This is related by the following equations:

$$\ddot{x} = J(q)\ddot{q} + \dot{J}(q)\dot{q} \tag{3}$$

$$\tau_{ext} = J(q)^T F_{ext} \tag{4}$$

For τ_d, this is equated to the desired force F_d and the Coriolis, centrifugal, and gravity torques to represent the impedance control equation without the non-linear effects.

$$\tau_d = J(q)^T F_d + C(q,\dot{q})\dot{q} + g(q) \tag{5}$$

By substituting 3, 4, and 5 into 2, the resulting simplified equation is the force and accelerations of the robot dynamics in Cartesian space:

$$\Lambda(q)(\ddot{x} - \dot{J}(q)\dot{q}) = F_d + F_{ext} \tag{6}$$

Where, $\Lambda(q)$ is the robot inertia matrix denoted as $(J(q)M(q)^{-1}J(q)^T)^{-1}$.

B. Impedance Dynamics

The impedance equation [3], which is the closed-loop form of the mass-spring-damper equation in Cartesian space is

$$\Lambda_d(\ddot{x}_d - \ddot{x}) + D_d(\dot{x}_d - \dot{x}) + K_d(x_d - x) = F_{ext} \tag{7}$$

Where, Λ_d is the desired inertia matrix, D_d is the desired diagonal damping matrix, and K_d is the desired diagonal

stiffness matrix. By substituting 6 into 7 by relating $\ddot{x}$. The finalized equation for the impedance controller is denoted as

$$F_d = \Lambda(q)\ddot{x}_d + \Lambda(q)\Lambda_d^{-1}[K_d(x_d - x) + D_d(\dot{x}_d - \dot{x})] + (\Lambda(q)\Lambda_d^{-1} - I)F_{ext} - \Lambda(q)\dot{J}(q)\dot{q} \tag{8}$$

The control command τ_d is computed based on 8 to control the robot manipulator. The compensation terms for Coriolis, centrifugal, and gravity torques are computed back as the final desired joint space torque commands.

$$\tau_d = J(q)^T F_d + C(q, \dot{q})\dot{q} + g(q) \tag{9}$$

III. Controller Design

In terms of practical implementation of the controller some modelling assumptions are made to simplify the controller. The manipulator will be controlled using desired positions hence, $\dot{x} = \ddot{x} = 0$. The motion is quasi-static assuming low joint accelerations $\dot{J}(q)\dot{q} = 0$. The inertia matrices are the identity to decouple the inertial behaviour from joint torques exploding to infinity and reduce computation $\Lambda(q) = \Lambda_d = I$ [3]. The simplified impedance control equation becomes

$$F_d = K_d(x_d - x) - D_d\dot{x} \tag{10}$$

As for passivity, since the stiffness and damping matrix are fixed and positive definite, the controller is inherently passive. This means that neither terms generate energy (spring stores potential energy and damping decays energy). As for stability to prevent overshoot and oscillations, a critical damping of 1 is chosen and can be defined in the follow equation from [3]

$$D_d(q) = A(q)D_{\dot{\zeta}}K_{d1} + K_{d1}D_{\zeta}A(q) \tag{11}$$

Where $A(q)A(q) = \Lambda(q)$, $K_{d1}K_{d1} = K_d$, D_ζ is the diagonal matrix of damping ratios $0 \leq \zeta_i \leq 1$ (0 is undamped and 1 is critically damped). For a damping ratio of 1 and an identity inertia matrix this simplfies to

$$D_d(q) = 2\sqrt{K_d} \tag{12}$$

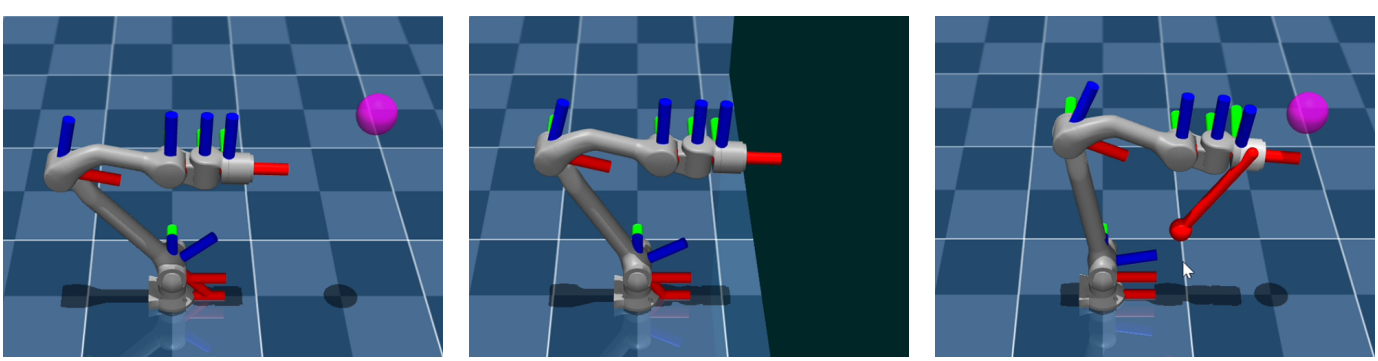
IV. Experiments

After deriving the simplified Cartesian impedance equation 10, the purpose of the experiment is to evaluate the performance and behaviour of the manipulator under varying stiffness and external forces. To evaluate the stiffness sensitivity, the chosen values for K_d are $50N/m$, $150N/m$, and $300N/m$ along the diagonal matrix. This has been experimentally chosen based on what is representative of sluggish to reactive tracking. The simulation environment is set up in MuJoCo and defines different environmental conditions, as shown in 1a, 1b, and 1c. This is to represent the tracking behaviour of the desired position with no disturbances, the interaction forces against a stiff environment, and the disturbance rejection/compliance from simulated human interaction. The desired set point for all cases is located at $< 0.4, 0.0, 0.3 >$ with respect to the world origin. The recorded data for the runs is the end-effector position and external force readings from MuJoCo over a 30 second run.

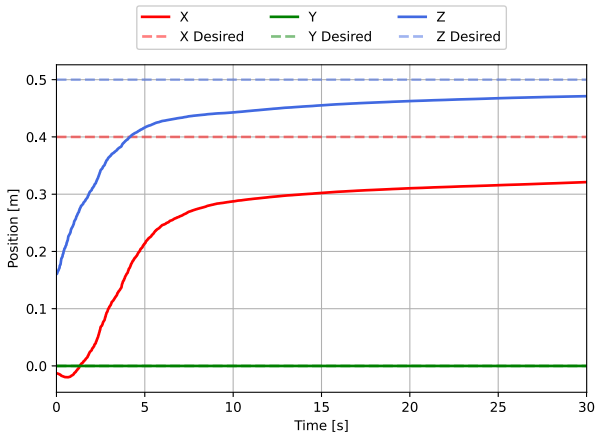
V. Results

This section summarizes the main behaviours of the manipulator under various stiffness and environmental conditions, the full results can be found in Appendix A. The results in the free motion experiment for the translational positions show that for increasing stiffness the end-effector reaches the desired position in a shorter time with a lower final error after the run. Based on the desired set point the Y-axis profile is constant as the goal is only in the X-Z plane. This can be seen in 2a and 2b showing that higher stiffness yields a greater attraction force to the desired position mitigating error in a smaller time horizon. As for the force profile as no external forces were imposed the profiles also represent this with a constant force of zero during the run.

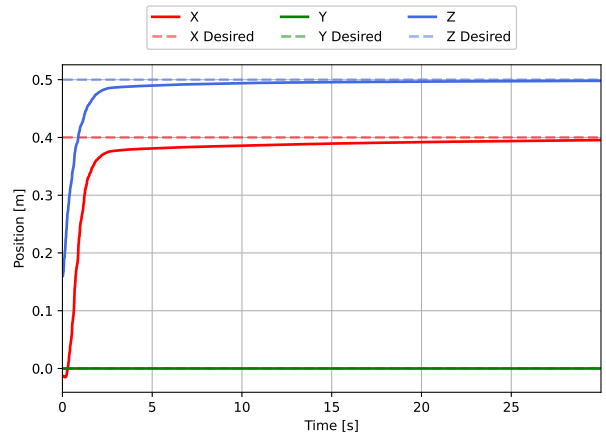
For the wall contact results, the position profile shows a ramp up to a constant steady state error, this is due to the wall intercepting the path of the manipulator never allowing it to reach the desired position. The ramping profile shows that for greater stiffness it collides with the wall at a higher rate. This can be better shown in the force profiles for 2c and 2d, the negative forces are the reaction forces from the wall and is characterized by a large impact force which drops down significantly. For each stiffness the maximum forces are around $-15N$, $-95N$, and $-220N$ respectively. This demonstrates that greater stiffness yields higher spring forces leading to greater impacts on stiff environment, demonstrating lower compliance at higher stiffness. The noise generated for the higher stiffness profiles are likely artifacts from the



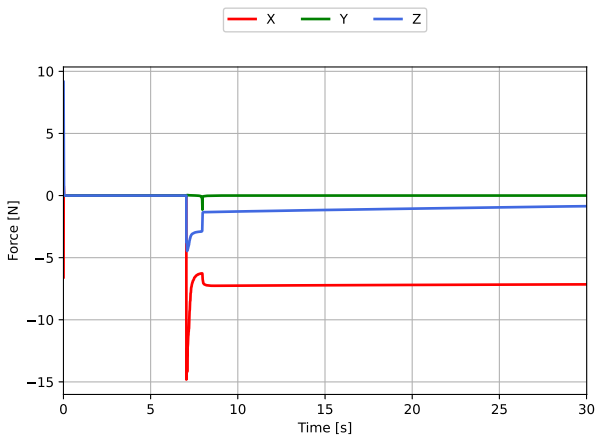
(a) Free motion (b) Wall contact (c) External disturbance



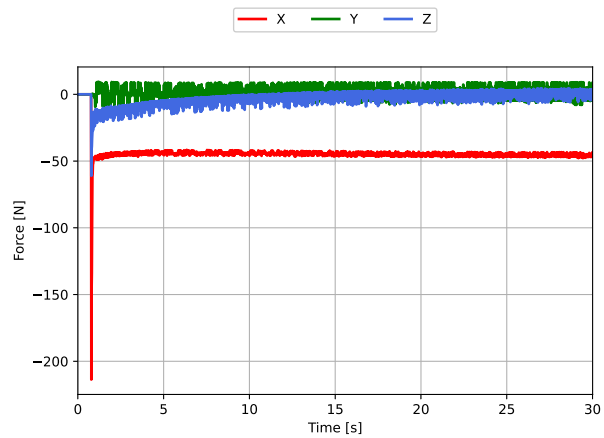
(a) Free motion position ($K_d = 50$)



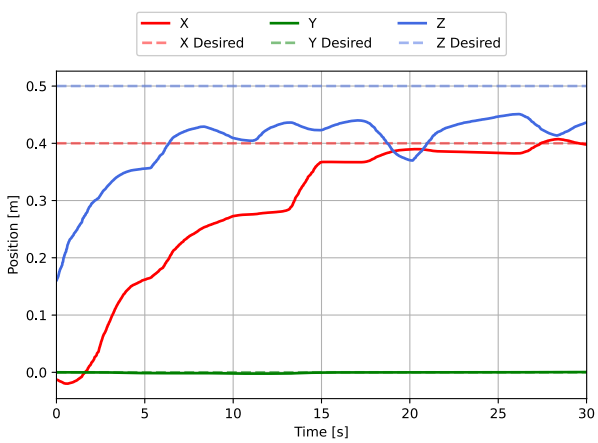
(b) Free motion position ($K_d = 300$)



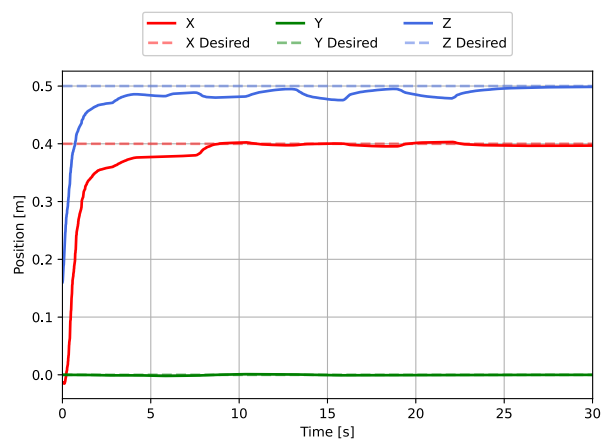
(c) Wall contact force ($K_d = 50$)



(d) Wall contact force ($K_d = 300$)



(e) External disturbance position ($K_d = 50$)



(f) External disturbance position ($K_d = 300$)

simulator as its detecting multiple collisions at different parts of the end-effector touching the wall, this is amplified as the applied force increases.

As for the external disturbance results the force profiles characterize similar tracking behaviour in the initial free motion experiment with added disturbance rejection. Figures 2e and 2f show that with increased stiffness the manipulator is less prone to external disturbances, resulting in less positional error for similar applied forces. By increasing the spring force any contribution from external disturbances become diminished resulting in lower compliance. The force profiles displayed in the Appendix A show the external forces applied

to the end-effector from the simulated human interaction, the variance of each profile for varying stiffness is due to human error (unable to re-create exact external forces) but the results are still meaningful in showing disturbance rejection/compliance.

VI. Discussion

Overall, the results demonstrate expected performance for each of the tests, with free motion it's expected that higher stiffness will improve accuracy but requires more joint effort. This couples well with how it performs against a stiff environment which means that although greater stiffness aids in reducing error the impact forces will also increase which can result in damage of the manipulator and/or obstacles/humans it collides with. This requires a balance between performance and safety applying reasonable constraints such as force limiting in the workplace, suitable for the impedance controller. For the external disturbance or simulated human interactions, the results demonstrate how compliant the manipulator is. Lower stiffness makes it easier to comply to human motion, in which the opposite is true that higher spring forces promote disturbance rejection. This would also be another design requirement considering if the person needs to move the end-effector for hand guiding tasks or not. In terms of stability the profiles across the runs do not show oscillatory behaviour (excluding the force profiles for the wall contact experiment as discussed in V), this is due to the damping selection ensuring the energy is decaying at critical damping, making it a suitable design choice to prevent overshoot or lagging behaviour.

The impedance controller are used in CoBots for human-aware force limiting and compliance, this can be used to promote safety in design but does not mean the controller itself is inherently safe. For high stiffness this can simulate position-based control as any external forces will be insignificant to the end-effector. This can yield to dangerous behaviours especially in collisions with people, hence ISO collaborative modes such as power for and limiting (PFL) are relevant to constrain the desired forces from the controller to the required safety range reducing risk of harmful collisions. Another relevant ISO collaborative mode is the hand guiding mode as by treating the desired position as the current position the spring forces are zeroed leaving only damping behaviour allowing for re-positioning of the end-effector. In terms of speed and separation monitoring (SSM) this is not within the scope of capabilities of the controller as this requires monitoring without contact through vision or other distance measuring techniques.

VII. Conclusion

This report covers the derivation and implementation of the impedance controller to evaluate its performance in a simulated environment for varying stiffness and environment conditions (free motion, wall contact, and external disturbance). The derivations show that by relating the Cartesian accelerations between the robot dynamic model and impedance equations the end-effector can virtually simulate desired mass-spring-damper. This results in compliant behaviour with the environment and objects it interacts with. There is a balance between performance and safety and is considered on the application level. This relates to industrial settings with different collaboration modes for safety from PFL to hand guiding. This work aimed to develop understanding and intuition behind force-based control for safety applications to characterize key design considerations of the controller.

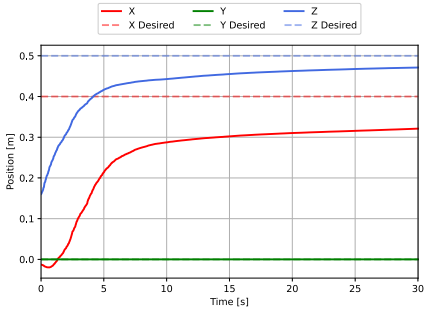
References

[1] E. Montini, F. Daniele, L. Agbomemewa, M. Confalonieri, V. Cutrona, A. Bettoni, P. Rocco, and A. Ferrario, "Collaborative robotics: a survey from literature and practitioners perspectives," *Journal of Intelligent & Robotic Systems*, vol. 110, no. 3, p. 117, 2024.

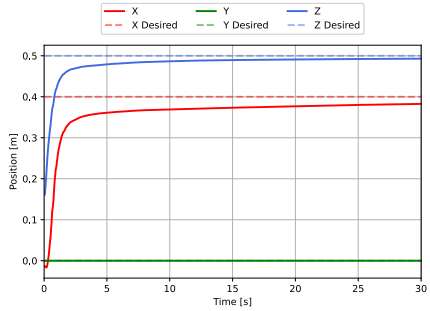
[2] F. Ferraguti, C. Secchi, and C. Fantuzzi, "A tank-based approach to impedance control with variable stiffness," in *2013 IEEE international conference on robotics and automation*. IEEE, 2013, pp. 4948–4953.

[3] A. Albu-Schaffer, C. Ott, U. Frese, and G. Hirzinger, "Cartesian impedance control of redundant robots: Recent results with the dlr-light-weight-arms," in *2003 IEEE International conference on robotics and automation (Cat. No. 03CH37422)*, vol. 3. IEEE, 2003, pp. 3704–3709.

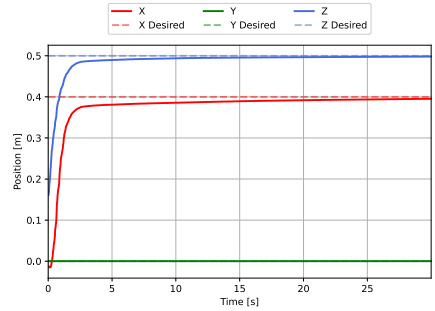
Appendix



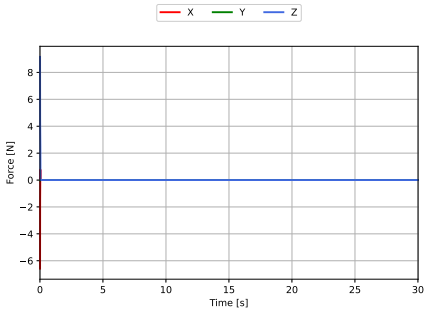
(a) Free motion position ($K_d = 50$)



(b) Free motion position ($K_d = 150$)



(c) Free motion position ($K_d = 300$)



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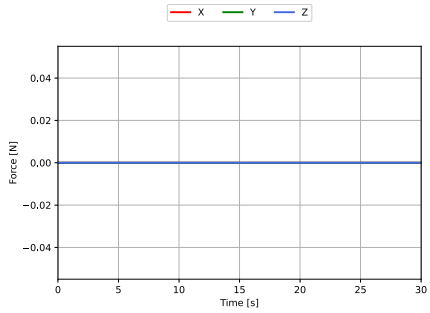

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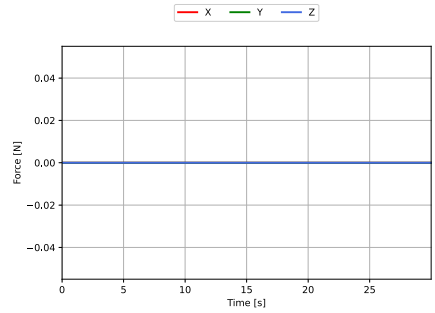
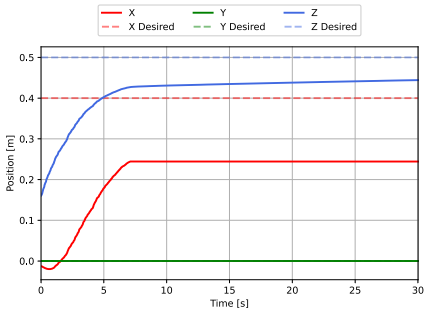
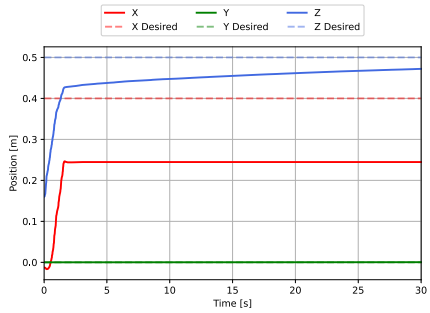


(f) Free motion force ($K_d = 300$)



(g) Wall contact position ($K_d = 50$)



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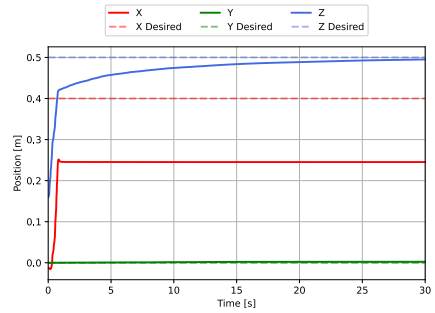
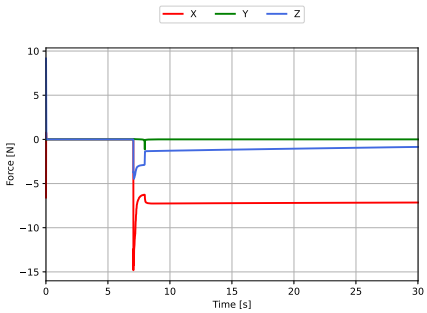
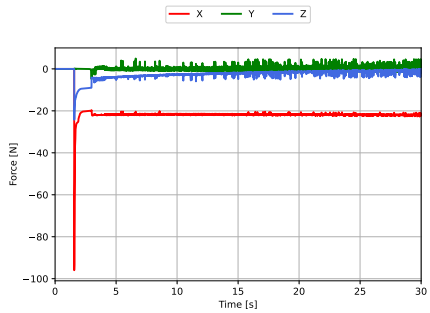


(i) Wall contact position ($K_d = 300$)



(j) Wall contact force ($K_d = 50$)



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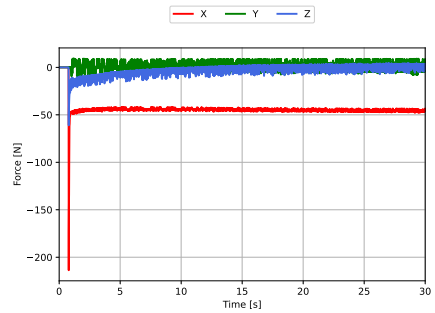
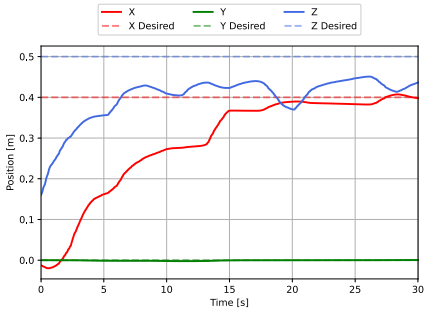
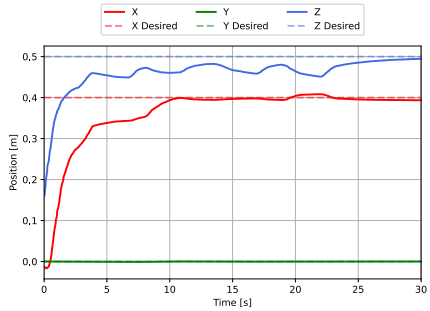


(l) Wall contact force ($K_d = 300$)



(a) External disturbance position ($K_d = 50$)



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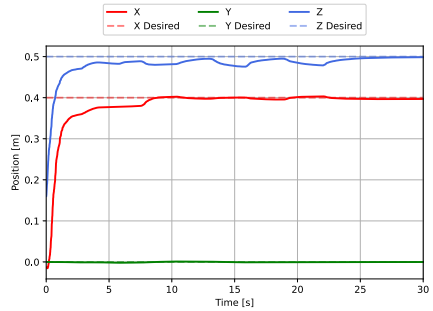

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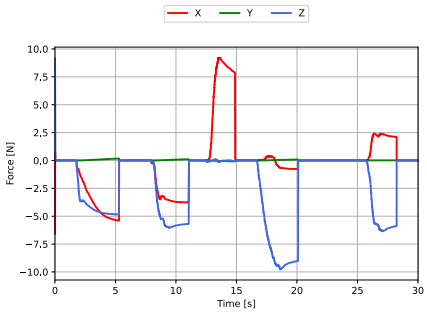

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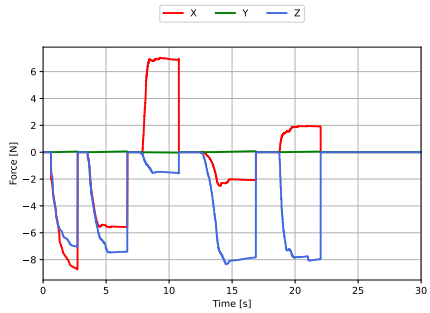

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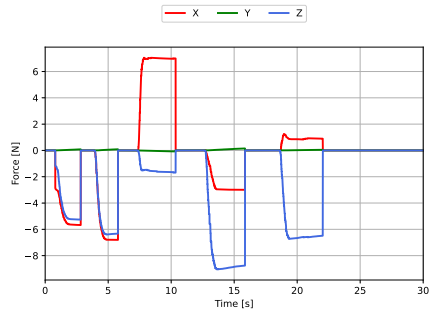

(f) External disturbance force ($K_d = 300$)